E Supplementary theorems

This appendix contains theorems that support the main results, providing additional context or being pre-requisite for the proofs in appendix F. We use the formulation of section 4, where $1 \leq \{t, t'\} \leq T$ are indices over tokens, $x \in \mathbb{R}^{N_x}$ is the message receiving embedding, $\{y_t \in \mathbb{R}^{N_y}\}$ are the message senders, $w_t = x^T W_{QK} y_t$, and $a_t = \texttt{softmax}_t w_t$ is the attention distribution

Theorem 9. *Shifting attention scores w_t by a constant offset does not affect the attention distribution. Therefore attention is fully determined by differences in scores.*

Proof. Applying the shift $w_t \xrightarrow[\text{offset w}]{} w_t + \delta w \; \forall \, t$ with fixed δw, we have

$$\begin{aligned} a_t \;&=\; \frac{e^{w_t}}{\sum_{t'} e^{w_{t'}}} \\ &\xrightarrow[\text{offset w}]{} \frac{e^{\delta w} e^{w_t}}{\sum_{t'} e^{\delta w} e^{w_{t'}}} \;=\; \frac{e^{\delta w}}{e^{\delta w}} \frac{e^{w_t}}{\sum_{t'} e^{w_{t'}}} \;=\; 1 \cdot a_t \;=\; a_t \end{aligned} \tag{18}$$

Alternatively we may write

$$a_t \;=\; \frac{e^{w_t}}{\sum_{t'} e^{w_{t'}}} \;=\; \frac{e^{w_t}}{e^{w_{t'}} \sum_{t'} e^{w_{t'} - w_t}} \;=\; \frac{1}{\sum_{t'} e^{w_{t'} - w_t}} \tag{19}$$

where $(w_{t'} + \delta t) - (w_t + \delta t) = w_{t'} - w_t$.

Theorem 10. *Multiplying attention scores by a positive factor changes the inverse-temperature of the attention distribution, modulating its sparsity (low temperature = less entropy = more sparse). Corollary: In the sparse limit, attention is fully determined by the order of w_t.*

Proof. Applying the scaling $w_t \xrightarrow[\text{scale w}]{} \kappa w_t \; \forall \, t$ with fixed $\kappa > 0$, we have

$$\begin{aligned} \frac{e^{\kappa w_t}}{\sum_{t'} e^{\kappa w_{t'}}} \;&=\; \frac{1}{\sum_{t'} e^{\kappa (w_{t'} - w_t)}} \\ &\xrightarrow[\kappa \to 0]{} \frac{1}{\sum_{t'} e^0} \;=\; \frac{1}{T} \; \forall \, t && \text{[fully isotropic distribution]} \\ &\xrightarrow[\kappa \to \infty]{} \begin{cases} 1 & \text{for } t = \text{argmax}_{t'} w_{t'} \\ 0 & \forall \, t \neq \text{argmax}_{t'} w_{t'} \end{cases} && \text{[fully sparse distribution]} \end{aligned} \tag{20}$$

where the `argmax` operator is fully determined by the order of w_t.

Theorem 11. *In the `No-Norm` case, the attention distribution a_t is defined by the projection of y_t onto a fixed vector y_x for a given x. The length of y_x is an inverse-temperature parameter.*

Proof. Write $w_t = x^T W_{QK} y_t = (W_{QK}^T x)^T y_t \equiv y_x^T y_t$ where $y_x \triangleq W_{QK}^T x \in \mathbb{R}^{N_y}$, which is the dot-product between y_t and a fixed vector y_x on the row space of W_{QK}. Then, re-writing in terms of the vector lengths and the enclosing angle $\theta_{y_t} = y_x \wedge y_t$, we have $w = |y_x||y_t| \cos \theta_{y_t}$. The factor $|y_x|$ is identical for all t, making it an inverse-temperature.

Theorem 12. *In the* `No-Norm` *case, bias parameters in the construction of query and key vectors are nullified by the* `softmax`*, or only contribute terms that may be recovered if x contains a constant direction.*

Proof. Consider a modification to the construction of query and key vectors that uses the affine transformations $q = W_Q x + b_Q$ and $k_t = W_K y_t + b_K$, with $W_Q \in \mathbb{R}^{N_{qkv} \times N_x}$, $W_K \in \mathbb{R}^{N_{qkv} \times N_y}$, $W_{QK} \triangleq W_Q^T W_K$, and $b_Q, b_K \in \mathbb{R}^{N_{qkv}}$. The dot-product attention scores are then:

$$
\begin{aligned}
w_t &= q^T k_t \\
&= (W_Q x + b_Q)^T (W_K y_t + b_K) \\
&= x^T W_{QK} y_t \;+\; ({W_Q}^T b_K)^T x \;+\; ({W_K}^T b_Q)^T y_t \;+\; b_Q^T b_K \\
w_t \;+\; const &= x^T W_{QK} y_t \;+\; ({W_Q}^T b_K)^T x \;+\; ({W_K}^T b_Q)^T y_t \\
&\triangleq x^T W_{QK} y_t \;+\; \rho_x^T x \;+\; \rho_y^T y_t \qquad \rightarrow\; \rho_x^T x = const \text{ given } x \rightarrow \\
&= x^T W_{QK} y_t \;+\; \rho_y^T y_t \qquad \rightarrow\; W_{QK} \triangleq \Omega^T \Lambda \Sigma \text{ via SVD } \rightarrow \\
&= x^T \Omega^T \Lambda \Sigma y_t \;+\; \rho_y^T y_t \qquad \rightarrow\; x' \triangleq \Omega x,\;\; y_t' \triangleq \Sigma y_t \;\rightarrow \\
&= {x'}^T \Lambda y_t' \;+\; \rho_y^T y_t
\end{aligned}
\tag{21}
$$

After expanding the terms, we find an additive constant $b_Q^T b_K$, and move this onto the LHS. Theorem 9 states that this has no impact on the output of the `softmax` operator. We identify $\rho_x \triangleq W_Q^T b_k$ and $\rho_y \triangleq W_K^T b_q$ as vectors on the **row-spaces of W_Q and W_K respectively**, defined as linear maps of the special directions b_K and b_Q. Since x is constant for each `softmax`, $\rho_x^T x$ is constant, and we absorb it into the LHS. We perform the singular value decomposition $W_{QK} \triangleq \Omega^T \Lambda \Sigma$ where $\{\Omega \in \mathbb{R}^{N_x \times N_x},\; \Sigma \in \mathbb{R}^{N_y \times N_y}\}$ are orthonormal matrices and $\Lambda \in \mathbb{R}^{N_x \times N_y}$ is a diagonal matrix of positive-semidefinite singular values with maximum rank $\min(N_x, N_y, N_{qkv})$. Orthonormal matrices apply a basis change to the embedding space using rotations and reflections. We write the transformed embeddings as $x' \triangleq \Omega x$ and $y_t' \triangleq \Sigma y_t$. The dot-product then has two terms:

1. ${x'}^T \Lambda y_t' = \sum_i \Lambda_{ii} x_i' y_{ti}'$ sculpts the attention distribution according to *pairwise relationships* between embeddings. We can say that $\{\Omega, \Sigma\}$ align the bases of x and y_t, mapping them onto a common orthonormal coordinate system. Λ_{ii} then assigns an importance weight to each coordinate i, determining the contribution of $x_i' y_{ti}'$.
2. $\rho_y^T y$ means "token t sends to all receivers when $y_t \parallel \rho_y$", where ρ_y must be a vector on the row-space of W_K. This may be recovered in the expansion of ${x'}^T \Lambda y_t'$ if there exists a direction i for which $x_i' = const$.

F Proofs of theorems in the main text

This appendix provides proofs for the theorems presented in section 5-6.

Theorem 1. *`No-Norm`: If two heads with finite non-zero temperature attend to different semantic subspaces, the subspaces must be linearly independent $\mathbb{S}_{\alpha}^{N_\alpha} \equiv \mathbb{R}^{N_\alpha}$. Corollary: W_{QK} is a low-rank matrix with (left and right) null-spaces that span all non-attended information.*

Proof. Let θ_A and θ_B be co-ordinates for the subspaces of x attended to by heads A and B respectively, and ϕ be all other information. Let $\theta_A \perp \theta_B \perp \phi$ and $x \perp y_t$, where $\perp$ denotes independence. Without loss of generality, write

$$x(\theta_A, \theta_B, \phi) \;=\; x_A(\theta_A) \;+\; x_B(\theta_B) \;+\; x_{other}(\theta_A, \theta_B, \phi) \tag{22}$$

Then write

$$\begin{aligned} w_t^{(A)}(\theta_A) \;&=\; \left(W_{QK}^{(A)} y_t\right)^T x(\theta_A, \theta_B, \phi) \\ &=\; \left(W_{QK}^{(A)} y_t\right)^T x_A(\theta_A) \;+\; \left(W_{QK}^{(A)} y_t\right)^T x_B(\theta_B) \;+\; \left(W_{QK}^{(A)} y_t\right)^T x_{other}(\theta_A, \theta_B, \phi) \end{aligned} \tag{23}$$

which requires $\left(W_{QK}^{(A)} y_t\right)^T x_B(\theta_B) = 0$ and $\left(W_{QK}^{(A)} y_t\right)^T x_{other}(\theta_A, \theta_B, \phi) = 0$, since any cancellation between the two terms must be independent of θ_A, ϕ and so can be absorbed entirely into the function $x_B(\theta_B)$. This means that $x_B(\theta_B)$ and $x_{other}(\theta_A, \theta_B, \phi)$ must both be orthogonal to $W_{QK}^{(A)} y_t$, meaning that they reside on the *left null space* of $W_{QK}^{(A)}$, or are projected by ${W_{QK}^{(A)}}^T$ onto a null space of y_t.

Head A can only attend to θ_A if $x_A(\theta_A)$ it is not on either of these null spaces, meaning that $x_A(\theta_A)$ is linearly independent of $x_B(\theta_B)$ and $x_{other}(\theta_A, \theta_B, \phi)$. Likewise for head B

$$\begin{aligned} w_t^{(B)}(\theta_B) \;&=\; \left(W_{QK}^{(B)} y_t\right)^T x(\theta_A, \theta_B, \phi) \\ &=\; \left(W_{QK}^{(B)} y_t\right)^T x_A(\theta_A) \;+\; \left(W_{QK}^{(B)} y_t\right)^T x_B(\theta_B) \;+\; \left(W_{QK}^{(B)} y_t\right)^T x_{other}(\theta_A, \theta_B, \phi) \end{aligned} \tag{24}$$

requires that $x_B(\theta_B)$ is linearly independent of both $x_A(\theta_A)$ and $x_{other}(\theta_A, \theta_B, \phi)$. Since x_{other} resides on both null spaces, it is linearly independent of both $x_A(\theta_A)$ and $x_B(\theta_B)$, and may be seen as a third subspace that passes information through to subsequent layers.

We can also write $w_t = \left(W_{QK}^T x\right)^T y_t$, and so the same argument also holds for subspaces on y_t. In this case, non-attended subspaces are spanned by the *right null space* of W_{QK}.